

# 01: Intro Stats Review

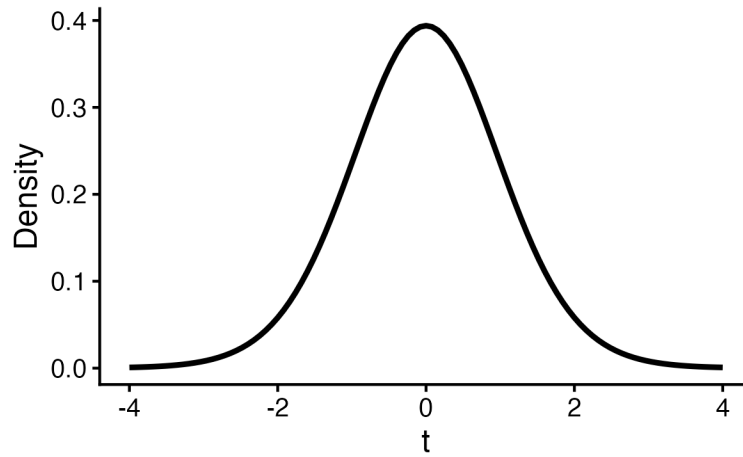
Stat 230 - Spring 2026

Researchers examined the relationship between sleep duration and simple reaction time performance in healthy young adults in a controlled laboratory setting.

Sixty-five college-aged participants (ages 18-25) were recruited and randomly assigned to sleep either 4 hours per night (**low sleep**) or 8 hours per night (**high sleep**) over a one-week period. Each morning, participants completed a computerized simple reaction time task where they responded to visual stimuli as quickly as possible, with reaction times measured in milliseconds. The researchers are interested in whether there is a difference in reaction time for the low sleep vs the high sleep group.

|                   | Sample size ( $n$ ) | Sample mean ( $\bar{x}$ ) | Sample SD ( $s$ ) |
|-------------------|---------------------|---------------------------|-------------------|
| <b>Low Sleep</b>  | 33                  | 390 ms                    | 45 ms             |
| <b>High Sleep</b> | 32                  | 360 ms                    | 40 ms             |

1. What is the null hypothesis for this test?
2. What is the alternative hypothesis for this test?
3. The researchers conduct a two-sample t-test and report a test statistic of  $t = 2.84$ . Below is a plot of the appropriate t-distribution for this test. Sketch how you would calculate the p-value for this situation.



4. How did you calculate this p-value in your last statistics course?
5. If the p-value you calculated was 0.02, what would you conclude about the null hypothesis?
6. Why did researchers randomly assign participants to a sleep condition?
7. Can you generalize these results to all young adults in the U.S.? Why or why not?